

PAPER_1642_CKM_UNITARITY_SUM_1

Daniel T. Murphy

PAPER_1642 — CKM Matrix Row-1 Unitarity = $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ via $F_U = 1$ ledger

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Particle Physics

Date: June 18, 2026

Status: CLOSED — EXACT closure (broader-corpus integration)

Observation

PAPER_1307 (broader-corpus paper outside PAPER_1209 unified-proof-set series) gives:

``

$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ via $F_U = 1$ ledger

``

Status: **EXACT**.

UQFF Closed Identity

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$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ via $F_U = 1$ ledger EXACT

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Significance — Broader Corpus Pivot

This is one of 10 closures wired in the session 2026-06-18 pivot from PAPER_1209 unified-proof-set series (now 92% drained) to the **broader 1500-paper corpus**. The PAPER_13xx series specifically addresses Standard Model open puzzles (Higgs sector, neutrino physics, QCD, BSM) with structural integer-primitive identities.

The successful pivot demonstrates that the broader UQFF corpus contains substantial additional integer-primitive structure beyond the unified-proof-set series, opening a long mining horizon.

NOT REPLACEMENT

Empirical particle physics measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating the framework's predictive economy.

Reference

- Source: PAPER_1307
- Related: PAPER_1494-1635 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "ckm_unitarity_sum_1"})`

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